

⚙️ RS003 Week 2 — English Worksheet

Topic: FPS, Title, and a Colour-Cycling Nebula · **Course:** Pygame Space Shooter · **Time:** about 45 minutes

This week you think about the code that makes last week's window feel **alive** — a **clock** that controls the frame rate (FPS), a nicer title bar, and a deep-space background that slowly shifts colour like a glowing nebula. You will read it, predict it, fix it, and explain it in your own words.

💬 Words to know: **clock**, **FPS (frames per second)**, **frame**, **tick**, **modulo (%)**

1 · Predict 🧠

Read each piece of code. Just from reading it, write what you think will happen. Do not run it.

```
clock = pygame.time.Clock()
FPS = 60
clock.tick(FPS)
```

What does `clock.tick(60)` do to the speed of the game loop?


```
frame_count = 0
r = (frame_count * 1) % 256
frame_count = frame_count + 1
```

`frame_count` goes up by 1 every loop. What does `% 256` do when it reaches 256?


```
r = (frame_count) % 256
g = 100
b = (200 - frame_count) % 256
screen.fill((r, g, b))
```

Red is climbing and blue is falling, while green stays still. Describe how the nebula colour changes as time passes.


```
pygame.display.set_caption("🌸 Space Shooter v2 - 60 FPS")
```

What part of the window will this line change?

2 · Spot the Bug 🐛

Each block below was meant to do something but is broken. Read it carefully and fix it on paper, then explain why the original was wrong.

Bug A — This game runs fine but your computer's fan gets loud and the CPU hits 100%.

```
running = True
while running:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False
    screen.fill((100, 100, 200))
    pygame.display.flip()
```

Hint: without a clock, the loop runs as fast as it possibly can.

Write the fixed code (show the loop):

Why was it wrong? Why does your fix work?

Bug B — The nebula should cycle colours, but the program crashes with an error about a colour value being too big.

```
r = frame_count
screen.fill((r, 100, 50))
```

Hint: colour numbers must stay between 0 and 255. `frame_count` keeps growing past 255.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The colour is supposed to keep changing, but it is stuck on one colour forever.

```
frame_count = 0
running = True
while running:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False
    r = frame_count % 256
    screen.fill((r, 100, 100))
    pygame.display.flip()
    clock.tick(60)
```

Hint: the frame counter never goes up.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Explain the Code

Read this working program line by line. Do not run it. Answer the questions using only what you can see.

```
import pygame
pygame.init()

WIDTH, HEIGHT = 800, 600
screen = pygame.display.set_mode((WIDTH, HEIGHT))
pygame.display.set_caption("🌸 Space Shooter v2 – 60 FPS")

clock = pygame.time.Clock()
FPS = 60
frame_count = 0

running = True
while running:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False

    r = (frame_count) % 256
    g = 100
    b = (200 - frame_count) % 256

    screen.fill((r, g, b))
    pygame.display.flip()

    clock.tick(FPS)
    frame_count = frame_count + 1

pygame.quit()
```

Which line decides how many frames happen each second, and what number is it set to?

The green value `g` is always 100. The red and blue values change. Explain why green stays the same colour but red and blue do not.

What does `frame_count = frame_count + 1` do, and why does the nebula need it?

What makes the loop stop and the window close?

`clock.tick(FPS)` is near the bottom of the loop. In your own words, what would happen to the speed of the colours if you removed it?

4 · Explain Your Lesson Code 🎥

Now explain the code **you** wrote during today's lesson. Record a short phone video where you talk through your own nebula code. You can show it running too. Try to use these words: **clock**, **FPS**, **frame**, **tick**, **modulo**.

1. Show your background slowly changing colour.
2. Read your colour formula out loud and say what makes it change.
3. Say what FPS you used and what `clock.tick` does for your computer.
4. Point to the line that makes the colour keep moving.

Write what you will say in your video. Plan it here before you record.

Submit ☒

Send this worksheet + a video explaining your lesson code to teacher on KakaoTalk.